

MASTER PLAN

TradePilot

v6

THE MACHINE

From Paper Trading to Market Domination

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1. Where We Are

TradePilot has evolved across four major versions, each adding a critical capability. Here is where things stand today.

VERSION	WHAT IT DOES	LINES	STATUS
v3	ML classification, regime-aware	~1,500	Retired
v4	7-signal composite scorer, long-only	~2,100	Live (paper)
v5	Multi-pool + shorts + regime + risk	~4,800	Live (paper)
v5.2	F&O options experiment	~1,300	Live (paper)
v6	The Machine	~15,000 est	Planning

2-Day Paper Trading Results

v4: Rs -19,279 – Day 1 bear: -30K, Day 2 bull: +11K

v5: Rs +38,986 – Day 1: -1.5K, Day 2: +40K

v5 proved the thesis: regime detection + shorts + risk management = profitable.

2. v6 Vision: "The Machine"

v6 is NOT a bigger v5. It is a fundamentally different architecture.

- v5 = Rule-based system with ML assistance
- v6 = Multi-agent AI system with rule-based safety rails

Architecture Overview



3. The Reality Check

The OpenAI "\$9K to \$90K" story is cherry-picked backtests, not a real product.

What ACTUALLY Works at Top Quant Funds

- **Renaissance:** Signal processing + HMM + statistical arbitrage (Sharpe ~2.0)
- **Two Sigma:** Gradient boosting on alternative data (15-20% annual)
- **Citadel:** ML for execution optimization

No LLM makes trading decisions. The AI is infrastructure, not the decision-maker.

Our Realistic Targets

METRIC	TARGET	WORLD-CLASS?
Sharpe ratio	1.5 – 2.0	Yes (top 1% of hedge funds)
Annual return	25 – 40%	Yes (beats 99% of mutual funds)
Max drawdown	<15%	Yes (institutional grade)
Win rate	60%+	Yes (with proper R:R)
Profit ratio	80% days profitable	Ambitious but achievable

4. Six Strategy Layers

v6 deploys ALL six simultaneously. Each layer is independent but cross-correlated via the Risk Agent.

Layer 1: Technical Momentum/Reversion EVOLVED

- LightGBM + TFT + LSTM ensemble on 500 stocks
- Dynamic blend: momentum in BULL, mean-reversion in BEAR
- Intraday (ORB, VWAP) + Swing (3-7d) + Positional (2-4wk)

Expected Sharpe: 1.0 - 1.5

Layer 2: Sentiment Alpha NEW in v6

- Claude/Llama sentiment scoring of news headlines + earnings calls
- 74% directional accuracy per research (Sharpe 3.05 in papers)
- Sources: MoneyControl, Economic Times, BSE announcements
- Contrarian at extremes (panic sell = buy signal)

Expected Sharpe: 1.5 - 3.0 (supplementary)

Layer 3: Institutional Flow ENHANCED

- FII/DII daily net flow – 1-3 day lead indicator
- Delivery percentage spikes – institutional accumulation signal
- Insider buying clusters – 11.2% annual alpha
- Bulk/block deal following

Expected Sharpe: 0.8 - 1.3

Layer 4: Cross-Asset Signals

NEW in v6

- India 10Y bond yield changes → equity sector rotation
- USD/INR trend → FII flow prediction
- Crude oil spikes → negative for Indian equities
- Gold rally + equity fall → risk-off regime
- US market overnight → gap prediction

Expected Sharpe: 0.8 - 1.2 (overlay)

Layer 5: Options Premium Harvesting

EVOLVED

- Weekly expiry straddle/strangle selling on Nifty/BankNifty
- IV consistently exceeds realized vol by 20-30% in India
- Protective puts during BEAR regime
- Covered calls on profitable SWING holdings

Expected Sharpe: 1.0 - 2.0

Layer 6: Pairs/Statistical Arbitrage

NEW in v6

- Cointegrated pairs within sectors (e.g., TCS-INFY, HDFCBANK-ICICIBANK)
- Market-neutral: profit regardless of market direction
- 4.97% annual on Nifty 50 pairs (from research)

Expected Sharpe: 1.5 - 2.5 (market-neutral)

5. Technology Stack

Component	Tool	Why
Live execution	Zerodha Kite Connect (Rs 2K/mo)	Best reliability, Python SDK
Tick data	Kite WebSocket (3000 instruments)	Real-time, ~1sec latency
Historical data	QuestDB + DuckDB	ASOF JOIN, fastest ingestion
ML training	LightGBM + neuralforecast (TFT/LSTM)	Proven, M1 Mac compatible
Sentiment	Claude API + Llama 3.1 (Ollama)	Best accuracy + local fallback
Feature store	DuckDB + Parquet files	Zero infra cost, AES-256
Backtesting	vectorbt + custom event-driven	Speed + accuracy
Monitoring	Grafana + Telegram bot	Real-time dashboards + alerts
Server	AWS Mumbai (ap-south-1)	Low latency to NSE, SEBI compliant
CI/CD	GitHub Actions	Auto-deploy on push

6. Go-Live Checklist

STEP	WHAT	TIMELINE	STATUS
1	Complete 20-session paper validation	2 more weeks	In Progress
2	Open Zerodha API account (Rs 2K/month)	1 day	TODO
3	SEBI algo registration	2-4 weeks	TODO
4	Build execution agent (Kite API wrapper)	1 week	TODO
5	Shadow mode (log orders, don't execute)	1 week	TODO
6	Live with 1 lot (tiny size, real money)	2 weeks	TODO
7	Scale to full capital	After 1 month live	TODO

7. Implementation Phases

Phase 1: Foundation (Weeks 1-4) – "Make It Work"

- Complete 20-session v5 paper validation
- Open Shoonya account, download 1 year intraday data
- Retrain ML ensemble (LightGBM + TFT + LSTM) with intraday features
- Build cross-asset signal overlay (bonds, crude, USD/INR)
- Add delivery percentage to signal engine

Target: Sharpe > 1.0, 60%+ win rate on paper

Phase 2: Intelligence (Weeks 5-8) – "Make It Smart"

- LLM sentiment layer (Claude API for news scoring)
- Insider buying data from SEBI filings
- Pairs trading module (cointegrated Nifty 200 pairs)
- Enhanced options strategies (volatility selling)
- Multi-agent orchestrator (coordinate all signal agents)

Target: Sharpe > 1.5, 65%+ win rate

Phase 3: Execution (Weeks 9-12) – "Make It Real"

- Zerodha Kite API integration
- Smart order routing (limit for liquid, market for urgent)
- Shadow mode testing (2 weeks)
- SEBI algo registration
- Kill switch + audit trail
- AWS Mumbai deployment

Target: Live with 1 lot, no catastrophic bugs

Phase 4: Scale (Weeks 13-20) – "Make It Bigger"

- Scale from 1 lot to full capital
- Add Grafana real-time P&L dashboard
- Commodities module (gold, crude)
- Currency hedging (USD/INR)
- Tax optimization engine (STCG/LTCG awareness)
- Public API for future subscribers

Target: 25-40% annual return, Sharpe > 1.5

8. Risk Management

Non-Negotiable Rules

RULE	LIMIT	WHY
Daily max loss	2% of capital	Survive any single day
Weekly max loss	5% of capital	Survive any bad week
Monthly max loss	10% of capital	Kill switch trigger
Single position max	10% of capital	Diversification
Max leverage (F&O)	3x	Don't blow up
Max correlated positions	3 from same sector	Concentration risk
Kill switch	Auto at -2% daily	Hard stop, no override
Recovery ladder	25/50/75/100% over 15d	Don't revenge trade

9. Competitive Analysis

FEATURE	ZERODHA	GROWW	ANGEL	TRADEPILOT V6
Stock trading	Yes	Yes	Yes	Yes (via Kite)
F&O	Yes	No	Yes	Yes + AI strategies
Algo trading	Manual	No	Basic	Full AI-powered
Regime detection	No	No	No	HMM + 8 indicators
Short signals	Manual	No	No	Auto-generated
Risk management	Basic SL	Basic	Basic	5-tier breakers
Sentiment analysis	No	No	No	LLM-powered
Multi-strategy	No	No	No	6 layers
Cross-asset	No	No	No	Bonds, crude, FX
Paper trading	Streak	No	No	Built-in

10. Revenue Model

After proving the system works (6+ months live track record):

TIER	PRICE	WHAT THEY GET
Free	Rs 0	Market data, basic signals, paper trading
Pro	Rs 999/mo	Full v6 signals, regime alerts, Telegram bot
Algo	Rs 4,999/mo	API access, auto-execution, all strategies
Enterprise	Custom	White-label, custom strategies, dedicated support

Target: 1,000 paying users at avg Rs 2,000/month = **Rs 20L MRR**

11. Why We Win

- 1 **We built the system before raising money** – 6,600 lines of battle-tested code across 4 versions.
 - 2 **We have live paper trading data** – not backtests, real market decisions with real-time signals.
 - 3 **Multi-strategy diversification** – 6 independent layers, not dependent on any single approach failing or succeeding.
 - 4 **India-first** – FII/DII flows, delivery %, F&O expiry patterns are our unique edge that no US-built system has.
 - 5 **AI as infrastructure, not hype** – regime detection + risk management, not "GPT picks stocks".
 - 6 **Bootstrapped, profitable, then scale** – same DevPilot philosophy. Prove it works, then grow.
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TradePilot v6 – The Machine

Built by traders, for traders.

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